🏆 REDROB RANKER
Hackathon Submission Overview

AI Candidate Discovery System & Architecture Overview

2. Solution Overview

Our proposed solution, **Recruiter Brain (Redrob Ranker)**, is an AI-driven candidate discovery and ranking system. It utilizes a high-performance **Two-Stage Retrieval and Scoring Pipeline** designed to run fully locally on CPU hardware, avoiding any latency spikes, API costs, or rate limits.

Stage 1 filters out 97% of irrelevant candidate profiles in less than 28 seconds. Stage 2 evaluates the remaining top 3,000 candidates using dense semantic embeddings and composite scoring algorithms to rank the top 100.

Comparison of Candidate Discovery Models:

Aspect	Traditional Keyword Matchers	Recruiter Brain (Proposed)
Matching Style	Regex, raw keyword counts, and keyword occurrences	Context-aware dense semantic embeddings
Signals Evaluated	Plain text skill tags only	Work experience tenure, activity recency, response rates, and notice periods
Fraud Protection	Vulnerable to keyword-stuffing and fake resumes	Robust Honeypot detector engine penalizing scams by 95%

3. Job Description (JD) Understanding

The system parses the JD into structured filters: required skills (Python, Embeddings, Vector DBs, ML), preferred experience range (5 to 9 years), and geographical hub matches. By mapping candidate summaries to a dense vector space using SentenceTransformers, the solution matches candidates semantically (e.g. matching 'RAG and vector search' to 'Embeddings database' requirements) rather than looking for literal keywords.

4. Ranking Methodology & 5-Axis Score

Candidates are evaluated along five distinct weighted categories:

- **Axis 1: Semantic Fit (35% Weight):** Cosine similarity score between JD embedding and candidate resume summary.
- **Axis 2: Technical Skills (30% Weight):** Direct experience scoring with vector DBs, Python, shipped ranking systems, and ML production pipelines.
- **Axis 3: Career Trajectory (15% Weight):** Stability score evaluating tenure consistency, penalizing frequent job hoppers, and favoring product company experience.
- **Axis 4: Availability & Recency (15% Weight):** Prioritizing candidate activity status, open-to-work flags, high response rates, and short notice periods.
- **Axis 5: Location Alignment (5% Weight):** Checking candidates' location details against target Indian tech hubs, or checking relocation willingness.

5. Explainability & Data Validation

Our Reasoning Generator is completely fact-grounded and deterministic, translating structured profile features directly into human-readable bullet points. This prevents the hallucination risk inherent in LLM approaches. Suspicious profiles are scanned by our Honeypot Engine which detects expert-skill explosions (claiming 8+ skills with <5 yrs experience), role overlaps, and duration overflows. These profiles are penalized by a multiplier of 0.05 (effectively eliminating them from the shortlist).

6. End-to-End Workflow & Architecture

The workflow ingests the JD and Candidate profiles, filters the candidates in parallel, and scores the survivors using localized batch sentence encoding before writing the submission CSV. The architecture is modular, separating requirements parsing (jd_parser.py), feature extraction (feature_extractor.py), and evaluation (scorer.py) for testing and development safety.

7. Performance Metrics & Results

- **Total Execution Time (100K candidates):** 138 seconds (~2.3 minutes)
- **Stage 1 (Pre-filtering time):** 28 seconds
- **Stage 2 (Embeddings and scoring time):** 105 seconds
- **Compute Requirements:** CPU-only, local model execution (zero cloud API calls)

8. Technologies Used

- **Sentence-Transformers:** Provides CPU-efficient embeddings creation using the lightweight all-MiniLM-L6-v2 model.
- **Streamlit:** Powering the high-fidelity dark-themed SaaS-like presentation and sandbox interface.
- **Python Multiprocessing:** Enables running parallel candidate feature extraction calculations over all available CPU cores.
- **NumPy and Pandas:** Powering fast vector dot products, normalization, and data frame export.

9. Submission Assets

GitHub Repository Link: https://github.com/YogapraveenRavikumar2026/redrob-ranker

Presentation Slides: redrob_ranker_deck.pdf